Vector Calculus and Complex Analysis Homework 3

1. Evaluate the line integral $\int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r}$, where $\boldsymbol{F}(x, y) = \left\langle \frac{x}{x^2+y^2}, \frac{y}{x^2+y^2} \right\rangle$ and C is any positively oriented simple closed curve enclosing the origin.

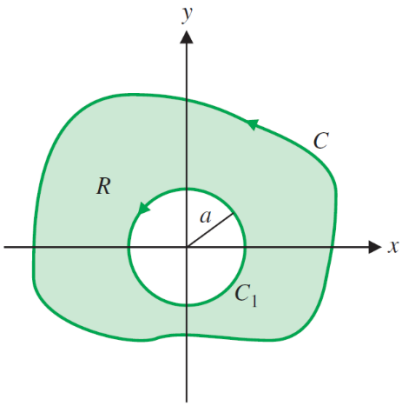
Solution

Let $M = \dfrac{x}{x^2+y^2}$ and $N = \dfrac{y}{x^2+y^2}$

$$\frac{\partial N}{\partial x} = -\frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial M}{\partial y} = -\frac{2xy}{(x^2+y^2)^2} = \frac{\partial N}{\partial x}$$

Define C, C_1 and R as



$$\oint_C \mathbf{F} \cdot d\mathbf{r} - \oint_{C_1} \mathbf{F} \cdot d\mathbf{r}$$

$$= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

$$= \iint_R 0 \, da = 0$$

Therefore, the line integral has the same value on all positively oriented simple closed curves containing the origin. So, we may assume that C is

$$x = \cos t, y = \sin t, 0 \le t \le 2\pi$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

$$= \oint_C \frac{x}{x^2+y^2}\,dx + \frac{y}{x^2+y^2}\,dy$$

$$= \int_0^{2\pi} [(\cos t)(-\sin t) + (\sin t)(\cos t)\,dt$$

$$= \int_0^{2\pi} 0\,dt = 0$$

Another method is $\boldsymbol{F}(x,y) = \left\langle \frac{x}{x^2+y^2}, \frac{y}{x^2+y^2} \right\rangle = \nabla\left(\frac{1}{2}\ln(x^2+y^2)\right) \Rightarrow \oint_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r} = 0$

2. Verify Green's Theorem for $\boldsymbol{F}(x,y) = \langle 0, 3x+2xy \rangle$ and the region $R = \{(x,y) : x^2+y^2 \le 4\}$

Solution

$$\oint_C (3x+2xy)dy = \int_0^{2\pi} (6\cos\theta + 8\sin\theta\cos\theta)2\cos\theta d\theta = \int_0^{2\pi} (12\cos^2\theta + 16\sin\theta\cos^2\theta)d\theta = 12\pi$$

$$\iint_R (3+2y)dA = 3\iint_R dA + 2\iint_R y dA = 12\pi + \int_0^{2\pi}\int_0^2 r^2\cos\theta dr d\theta = 12\pi + \left(\int_0^2 r^2 dr\right)\left(\int_0^{2\pi}\cos\theta d\theta\right) = 12\pi$$

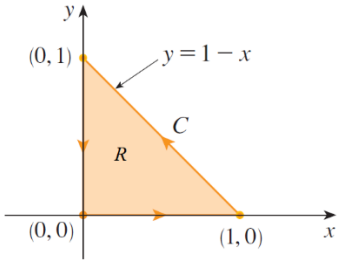
3. Verify Green's Theorem for $\boldsymbol{F}(x,y) = \langle y^2 - 2x, x^2 \rangle$ and $R = [0,1]\times[0,1]$.

Solution

(a) Bottom: $(0,0)$ to $(1,0)$.
$x = t, y = 0, 0 \le t \le 1$
$\oint_C (y^2-2x)\,dx + x^2\,dy = \int_0^1 -2t\,dt = -1$
Right: $(1,0)$ to $(1,1)$.
$x = 1, y = t, 0 \le t \le 1$
$\oint_C (y^2-2x)\,dx + x^2\,dy = \int_0^1 1\,dt = 1$
Top: $(1,1)$ to $(0,1)$.
$x = 1-t, y = 1, 0 \le t \le 1$
$\oint_C (y^2-2x)\,dx + x^2\,dy = \int_0^1 (1-2t)\,dt = 0$
Left: $(0,1)$ to $(0,0)$.
$x = 0, y = 1-t, 0 \le t \le 1$
$\oint_C (y^2-2x)\,dx + x^2\,dy = \int_0^1 0\,dt = 0$
Summing up the integrals over the four sub-paths, the line integral is 0.

(b) $M = y^2 - 2x, N = x^2$
$\oint_C M\,dx + N\,y$
$= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$
$= \iint_R [2x-2y]\,dA$
$= \int_0^1 \int_0^1 (2x-2y)\,dy\,dx$
$= \int_0^1 (2x-1)\,dx = 0$

4. Verify Green's Theorem for $\boldsymbol{F}(x,y)=\langle x^4, xy\rangle$ and the region R bounded by (0,0), (1,0) and (0,1).



Solution

$$\oint_C x^4dx+xydy=\int_{(0,0)\to(1,0)}x^4dx+xydy+\int_{(1,0)\to(0,1)}x^4dx+xydy+\int_{(0,1)\to(0,0)}x^4dx+xydy$$

$$=\int_0^1 x^4dx+\int_0^1(1-t)^4(-dt)+t(1-t)dt-\int_0^1 0(y)dy$$

$$=\frac{1}{5}-\frac{1}{5}+\frac{1}{2}-\frac{1}{3}=\frac{1}{6}$$

$$\iint_R\left(\frac{\partial F_2}{\partial x}-\frac{\partial F_1}{\partial y}\right)dA=\int_0^1\int_0^{1-x}ydydx=\frac{1}{2}\int_0^1(1-x)^2dx=\left.\frac{-(1-x)^3}{6}\right|_0^1=\frac{1}{6}$$

5. Calculate $\oint_C(\tan x-y^3)dx+(x^3-\sin y)dy$, where C is the circle $x^2+y^2=2$ oriented counterclockwise.

Solution

$$\oint_C F_1\,dx+F_2\,y$$
$$=\iint_R\left(\frac{\partial F_1}{\partial x}-\frac{\partial F_2}{\partial y}\right)\,dA$$
$$=\iint_R(3x^2+3y^2)\,dA$$
$$=\int_0^{2\pi}\int_0^{\sqrt{2}}3r^3\,d\mathbf{r}\,d\theta$$
$$=\int_0^{2\pi}3\,d\theta=6\pi$$